

PROJECT
TITLE



Next word prediction using NLP

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1. ABSTRACT

- .Next Word Prediction predicts the most probable word following a sequence.
 - .Used in messaging, search engines, writing assistants.
 - .Explores approaches, research gaps, and improved methods.
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2 INTRODUCTION



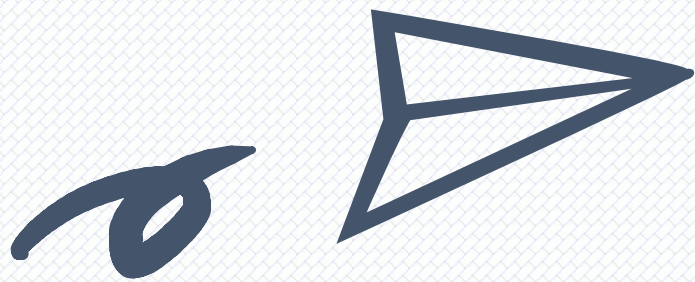
NLP enables machines to understand and generate human language.

Next Word Prediction improves typing efficiency and user experience.

Examples: Google Smart Compose, SwiftKey, Grammarly.

This presentation explores the evolution, methodologies, results, and future of next word prediction in NLP, from basic statistics to state-of-the-art models.

3 LITERATURE REVIEW



- .N-gram Models – simple but sparse data issues.**
- .Hidden Markov Models – probabilistic, sequential dependencies.**
- .Neural Networks (RNN, LSTM, GRU) – handle long-term dependencies.**
- .Transformers (BERT, GPT) – state-of-the-art contextual performance.**

4 RESEARCH GAP



Difficulty with long context understanding in traditional models.
Limited domain-specific adaptations.



Need for lightweight real-time models.
Bias and ethical issues remain challenges.

5 OBJECTIVE

Design contextually accurate prediction system.

.Reduce computation while maintaining accuracy.

.Explore transformer-based methods for adaptability.



.Model training workflows that transform raw text into intelligent prediction systems

.Identify promising research directions that will shape the next generation of language models

.Explore both traditional statistical methods and cutting-edge neural network techniques

6 EXISTING SYSTEM

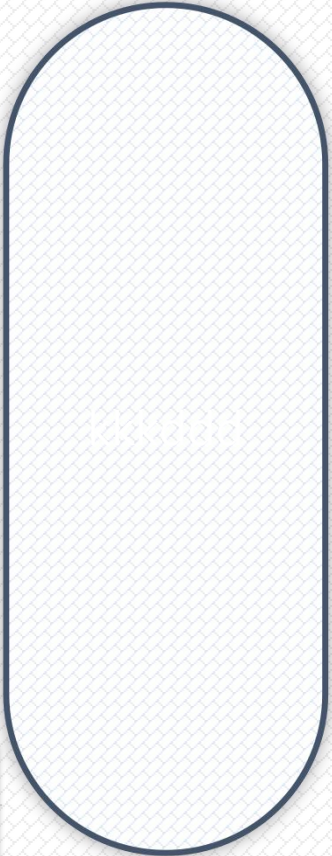
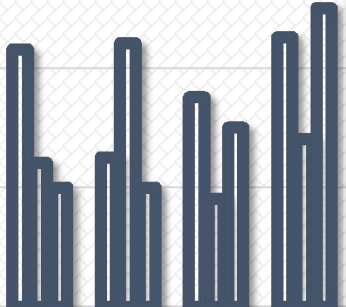
ENGINES

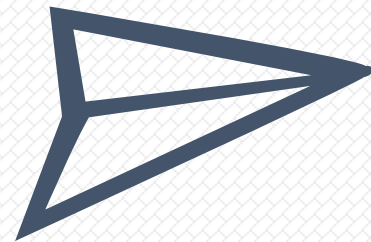
- .Google Gboard – deep learning for predictive typing.
- .Microsoft SwiftKey – adapts to user style.
- .Grammarly – contextual corrections and completions.
- .OpenAI GPT Models – advanced language generation.

GRAMS

- .Bigrams: $P(\text{word}|\text{previous word})$
- .Trigrams: $P(\text{word}|\text{previous 2 words})$
- .4-grams and beyond for extended context
- .N-gram models predict the next word based on the previous $N-1$ words

7. PROPOSED SYSTEM

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1. Hybrid model with lightweight transformers.
 2. Personalization to user writing style.
 3. Bias mitigation with fairness-aware training.
 4. Low-latency deployment for mobile/edge devices.



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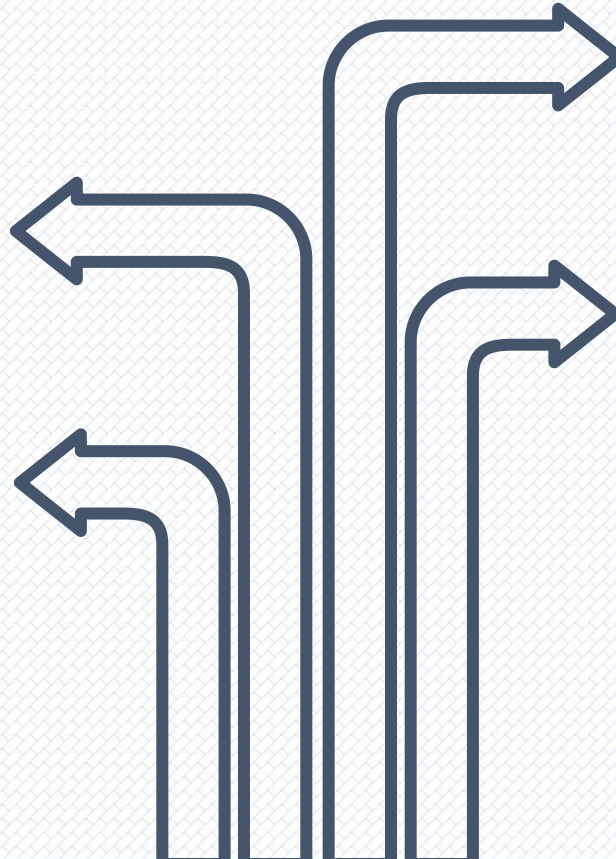
METHODOLOGIES

Data Collection –
Wikipedia, news, social
media datasets.

Preprocessing –
tokenization, stopword
removal, stemming.

Training & Fine-
tuning with
supervised learning.

Evaluation Metrics –
Perplexity, Accuracy,
BLEU score.



9 IMPLEMENTATION

Frameworks:
TensorFlow
/ PyTorch.

Model:
DistilGPT for
lightweight
prediction.

Testing:
Compare
accuracy with N-
gram and RNN.

Deployment:
mobile
integration.

Model
Architecture
Design:
Build
embedding layer



10. RESULTS

- .Transformers outperform N-gram and RNN models.**
- .Lower perplexity indicates improved language understanding.**
- .Hybrid fine-tuned models promising for real-time use.**
- .Split text into tokens and create n-gram sequences for training**
- .Implement early stopping and learning rate scheduling for optimal convergence.**



11 REFERENCES



1.Jurafsky & Martin – Speech and Language Processing.

2.Vaswani et al. – Attention is All You Need (2017).

3.Hochreiter & Schmidhuber – LSTM (1997).

4.Brown et al. – Language Models are Few-Shot Learners (2020).

12. FUTURE IMPLEMENTATION & REAL TIME APPLICATIONS

Future Implementation

1. Multilingual prediction for global adaptability.
2. Integration with speech-to-text systems.
3. On-device training for privacy-preserving personalization.

Real-Time Applications

- ✓ Text Messaging Apps – WhatsApp, Messenger.
 - ✓ Email Assistants – Gmail Smart Compose.
 - ✓ Accessibility Tools – faster communication for differently-abled.
 - ✓ Chatbots & Virtual Assistants – improved conversations.
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THANK YOU

